Decentralized Policy-Making and Market Distortions: Evidence from China's Drug Formulary Design*

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Abstract

When does decentralized policy-making get captured by local economic interests, and how large is the resulting distortion? We study this question in the design of public health insurance drug formularies in China, where provincial governments independently set drug coverage between 2000 and 2019 before the central government unified the formulary nationally. Using novel data on provincial insurance coverage for drugs linked to their producers, we show that the presence of a local producer raises a drug's probability of insurance coverage by four percentage points (10% of the mean), holding fixed disease, therapeutic class, and province-version factors. Favoritism is concentrated where officials face stronger career-promotion incentives, where they are natives of the province, and among state-owned and joint-venture firms. It also tilts coverage toward lower-quality drugs, suggesting welfare-relevant consequences. A simple framework decomposes the welfare difference between decentralized and centralized formularies and clarifies the conditions under which centralization is preferable despite the loss of local information. *JEL Codes:* H77, D72, D73, I13, L65.

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1 Introduction

A central question in public economics and political economy is at what level of government a given policy should be set. The classical reason for decentralization rests on local governments' superior knowledge of local conditions and preferences (Oates, 1999). The reason for centralization rests on inter-jurisdictional spillovers, equity in access, and—critically for this paper—the risk that local decision-makers are captured by local interest groups (Bardhan and Mookherjee, 2000; Ogawa and Wildasin, 2009). Whether decentralization or centralization is preferable in any given policy domain is therefore an empirical question, and the answer turns on the relative magnitudes of the information advantage and the capture distortion.

We study this question in the context of public health insurance drug formularies—the lists of pharmaceuticals reimbursed by social insurance programs—in China between 2000 and 2019. Drug formularies are a high-stakes policy instrument: they shape access to medicines for hundreds of millions of patients and direct hundreds of billions of dollars of pharmaceutical revenue. They also exist in essentially every country with public drug coverage, and the question of whether to set them nationally or sub-nationally is actively contested in Canada, Spain, Italy, and the United States, among others (Persaud et al., 2019; Tarricone et al., 2021). China's institutional setting is particularly informative for three reasons. First, between 2000 and 2019 the central government allowed each of 31 provinces to add drugs to its own list, generating direct, granular variation in policy choices. Second, in 2019 the central government reversed course and unified the formulary nationwide, citing concerns about local protectionism—providing a real-world policy decision for which our results speak directly. Third, rich firm-level data allow us to link coverage decisions to the economic interests of producers headquartered in each province.

We use this setting to ask three questions. First, do provincial governments use drug formulary discretion to favor producers based in their own province? Second, what mechanisms drive any such favoritism—private information, political incentives, or capture? Third, what are the welfare consequences relative to a centralized alternative?

To answer the first question, we assemble novel data on provincial drug formularies in 2009 and 2017—the two versions before the 2019 unification reform—linked to the universe of producers and a mapping from active ingredients to therapeutic classes and disease indications. We compare coverage rates for drugs with and without local producers, absorbing province-by-version-by-disease fixed effects. The identification comes from variation in local-producer presence across drugs treating the same disease in the same province and year, which absorbs demand-side factors such as local disease prevalence and regional treatment norms. The identifying assumption is that, conditional on these fixed effects, local-producer presence is uncorrelated with unobserved drug-by-province demand factors.

We find that the presence of a local firm raises the probability of inclusion by four percentage points relative to a sample mean of 41%—a 10% effect, robust across specifications. An event study of the 2017 revision confirms that inclusion is consequential: covered drugs see revenues rise by more than 50% within three years, with local firms capturing a disproportionate share. Provinces are favoring local producers in a setting where the favors translate into substantial rents.

To answer the second question—what drives this favoritism—we exploit variation in officials' incentives and origins. Local favoritism is significantly larger in provinces whose top officials face stronger career-promotion incentives, and larger where top officials are natives of the province they govern. Within local firms, the effect is concentrated in state-owned enterprises and joint ventures rather than private firms, and is larger for firms contributing more to provincial tax revenues and employment. Together these patterns are consistent with two well-known channels in the political economy literature: career-concerns motives operating through fiscal and employment outcomes (Li and Zhou, 2005), and capture by local interest groups (Jia and Nie, 2017).

To answer the third question—welfare—we link drugs to a national database of post-market quality reports, including manufacturing defects, severe adverse reactions, and safety warnings. Coverage is generally negatively correlated with reported quality problems, as one would expect from a benevolent designer. But this relationship is sharply attenuated, and sometimes reversed, for drugs with local producers. We also find no evidence that native officials are more responsive to quality information of local products than non-native officials, weighing against the interpretation that native officials possess a useful private signal about local products. Provincial discretion appears to skew coverage toward lower-quality local drugs—the cost side of the centralization trade-off is real and quantitatively meaningful.

We develop a simple conceptual framework to organize these findings. The framework decomposes the welfare difference between decentralized and centralized formularies into three components: a loss from neglecting heterogeneous preferences under uniform coverage, a loss from the central planner's incomplete information about local conditions, and a gain from removing the capture distortion. We then discuss the likely magnitudes of each component in light of the institutional context—in particular, the rapid consolidation of healthcare data systems in China, which is shrinking the information advantage of provincial governments, and the feasibility of centrally administered but province-specific formularies, which can mitigate the heterogeneous-preference loss. Together these arguments suggest a directional rationale for the 2019 unification reform.

Our paper contributes to three strands of literature. The first is the empirical literature on fiscal federalism and the capture of local governments. Existing work has documented the two sides of the federalism trade-off largely in isolation. On the information-advantage side, Huang et al. (2017) show that local governments' knowledge of local conditions improves outcomes in the management of state-owned enterprises. On the capture side, He, Wang and Zhang (2020) and He, Pan and Xie (2023) document weak enforcement of environmental regulation by local governments, where externalities make local accountability incomplete. Our contribution is to provide direct empirical evidence on the capture component of the federalism trade-off. The setting is unusually well-suited to this purpose: we observe the policy choice (drug-by-drug coverage in 31 provinces), the local economic interests at stake (firm-level producers linked to officials' incentives), and a welfare-relevant consequence (product quality)—a combination rarely available in a single empirical setting.

The second literature concerns local protectionism and government favoritism, particularly in developing countries. Young (2000) provides early evidence on local protectionism in China, and a growing body of subsequent work has documented the channels through which subnational governments favor firms within their jurisdictions: subsidies to local automobile manufacturers (Barwick, Cao and Li, 2021), discriminatory industrial policies (Bai, Tao and Tong, 2008), biased implementation of environmental regulation (Bai et al., 2021), judicial capture (Liu et al., 2023), and biased government procurement (Fang, Li and Wu, 2022). This literature has established that local favoritism is widespread, costly, and shaped by officials' political incentives (Li and Zhou, 2005; Jia and Nie, 2017). We add an under-explored channel: manipulation of public health insurance formularies.

Finally, we also connect to the healthcare literature on optimal formulary design. Most previous work pertains to the US health insurance markets and examines the strategic drug formulary design of insurance companies for risk selection (Geruso, Layton and Prinz, 2019; Lavetti and Simon, 2018). Additional studies examine how the establishment of drug insurance coverage may increase drug utilization and improve health outcomes, as observed in the US Medicare Part D (Duggan and Scott Morton, 2010) and Canada's compulsory health insurance (Wang et al., 2015). In the context of China, the previous literature documents that establishing the basic insurance program and drug formularies incentivizes innovation (Zhang and Nie, 2021). Chen and Han (2023) find that the formulary design is influenced by firms' political connections to government officials. We focus on the optimal drug formulary design, specially whether to vary it by province or adopt a national list. To our knowledge, this is the first empirical study of drug formulary design as a public-sector federalism question, and the first to link sub-national drug formulary decisions to product quality outcomes.

The outline of this paper is as follows. Section 2 provides institutional details. Section 3 describes the data and sample. Section 4 presents the empirical analysis. Section 5 presents the conceptual framework. Section 6 concludes.

2 Institutional Background

China's pharmaceutical market is one of the largest in the world. Drug expenditure increased from 221.12 billion CNY to 1,914.89 billion CNY between 2000 and 2018. Most drugs consumed are off-patent products from local firms, including domestic producers specializing in generic products and JVs producing and selling originator and patented drugs. The market is highly segmented, with over 5,000 domestic manufacturers (Kanavos, Mills and Zhang, 2019). Many of these manufacturers are small-scale, lack research and development (R&D) capacity, face high entry barriers, and primarily cater to local markets.

One potential driver of China's rapid growth in drug demand is the establishment of a basic health insurance program. In 2000, China introduced the "Basic Medical Insurance Drug Catalog", specifying covered drugs and coverage levels.[1] The formulary comprised two coverage tiers: Catalog A (full coverage) and Catalog B (10-20% cost-sharing). The central government set Catalog A nationally. However, for Catalog B drugs, due to the national formulary's lim-

[1] The insurance coverage status varied by active ingredient and dosage form (tablet, capsule, injections, etc). All products of a listed drug—branded or generics—were eligible for insurance coverage, though product availability varied by province.

ited coverage capacity and regional heterogeneity in disease prevalence, medication habits, and economic development, the central government set a core National Catalog B list and allowed provincial governments to add extra drugs.[2] Public health insurance budgets were administered sub-nationally. Hence, Provincial Catalog B variation granted provincial governments budgetary autonomy.

The formulary was revised in 2005, 2009, and 2017, with drugs added and removed across all tiers. In 2019, the central government began gradually unifying provincial formularies into a single national formulary by 2022.[3] This reform is part of efforts to shift social insurance administration from local to central control and addresses concerns about distorted incentives and lobbying in provincial formulary design. Provincial governments may manipulate formularies to favor local firms (Song, 2019); for example, a 2016 provincial government report stated that local products should be prioritized in formulary design.[4]

3 Data and Sample

We leverage the variation in provincial drug formularies across provinces before 2019 to empirically examine whether provincial formulary decision-making disproportionately benefits local firms. This section introduces our data sources and the methods used to construct the sample.

3.1 Basic Medical Insurance Drug Catalog Data

We compile the 2005, 2009, and 2017 versions of China's Basic Medical Insurance Drug Catalog (Catalog A, National Catalog B, and Provincial Catalog B) from government websites, consulting reports, and white papers, covering 31 provinces (listed in Appendix Table A2). In the formulary, observations are organized by active ingredient, dosage form, province, and version. We define a "drug" as an active ingredient-dosage form combination. Our analysis focuses exclusively on Western medicines, as traditional Chinese medicines have less clear classifications and indications, complicating demand measurement.

We identify 2,533 drugs ever added to Provincial Catalog B in 2005, 2009, or 2017 in any

[2]The number of provincial additions cannot exceed 15% of the National Catalog B.

[3]A separate formulary remained for traditional Mongolian and Tibetan medicines used by ethnic minority groups.

[4]See `https://www.hunan.gov.cn/xxgk/wjk/szbmwj/201607/20199245/files/ae74359e7bd94720953eceef36d52f0d.doc` (in Chinese).

province, expanding this to a drug-province-version panel. Our baseline analysis focuses on observations from the 2009 and 2017 formulary versions.[5] For each observation, we generate a dummy variable indicating Provincial Catalog B coverage.

We map drugs to WHO Anatomical Therapeutic Chemical (ATC) classification codes and then to WHO disease categories, following the reconciliation procedure of Costinot et al. (2019).[6] Each drug may map to multiple diseases, yielding 223 disease combination groups, which we use to control for disease prevalence and demand heterogeneity.

Figure 1 shows drug coverage across formulary versions. Catalog A (blue bars) and National Catalog B (brown) are uniform across provinces and expanding over time. Provincial Catalog B (orange) varies by province: the mean number of drugs ranges from 200 to 500, with error bars indicating the 10th-90th percentile range. We also find substantial variation in the fraction of drugs covered within each disease category across provinces (see Appendix Figure A2).

3.2 Firm Location, Ownership, and Portfolio Data

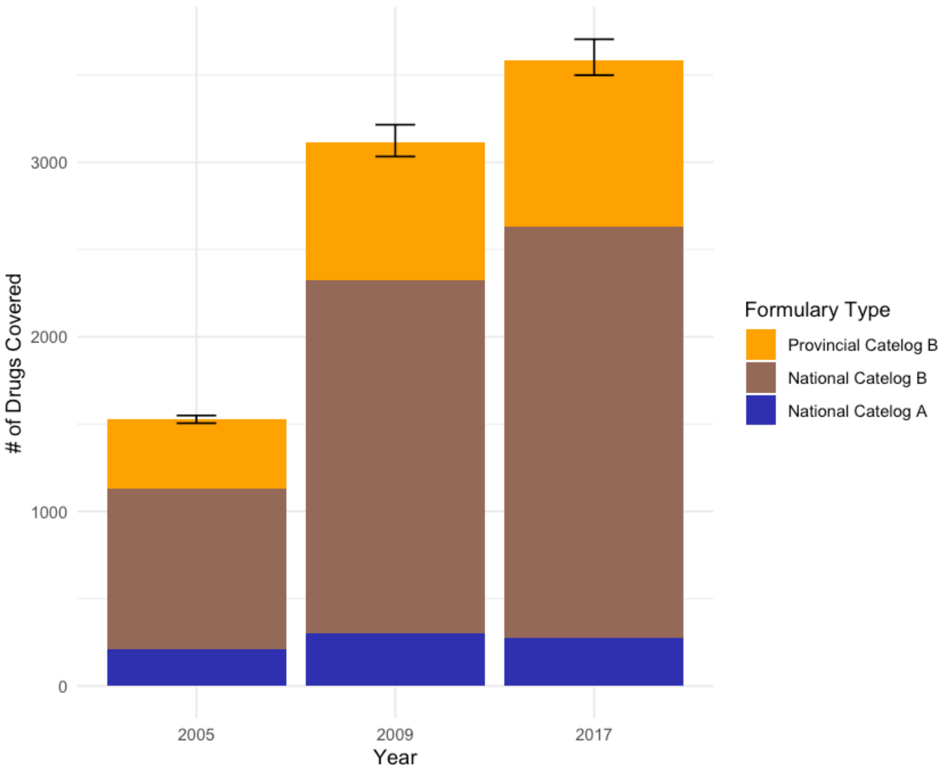
We identify firms' product lists using drug registration data from the National Medical Products Administration, which includes initial registration dates, and reclassify dosage forms to ensure consistency with the formulary data. Firm locations are drawn from China's Industrial and Commercial Registered Enterprises Database, supplemented by manual internet searches where data are missing.

We match each drug to its producing firms and generate indicators for local firm presence and count by drug-province-version in the year preceding the formulary. Appendix Figure A1 shows that 20–40% of covered drugs have local producers in most provinces, consistent with China's pharmaceutical industry comprising numerous small, locally serving manufacturers. We classify firms into four ownership categories: SOEs, JVs, domestic private firms, and foreign firms.

[5] Due to missing data on some variables used for the mechanism analysis for 2005, the baseline sample excludes observations from that year. However, the results remain consistent if the 2005 data are included (see Appendix Table A3).

[6] The WHO ATC system classifies active ingredients by anatomical, chemical, pharmacological, and therapeutic properties across five hierarchical levels, where Levels 3 and 4 denote close substitutes (Dubois and Lasio, 2018).

Figure 1: The Number of Drugs Listed in Each Formulary across Provinces



Notes: This figure shows drug coverage across the 2005, 2009, and 2017 formulary versions. Blue and brown bars indicate the number of drugs in Catalog A and National Catalog B, respectively. Orange bars show the mean number of drugs in Provincial Catalog B across provinces, with error bars representing the 10th-90th percentile ranges.

3.3 Drug Sales Data

We collect drug sales data from a consulting firm. The data records quarterly sales revenue and quantity for Western drug products in more than 500 public hospitals in 20 cities from 2013 to 2020.[7] The data include detailed product information: product name, production firm, main ingredients, route of administration, dosage form, strength, and package size.

We aggregate the sales data to the active ingredient–dosage form–city–quarter level, summing revenues across package sizes, weights, and brands. For each observation, we calculate local firms' sales revenue share and construct dummies indicating coverage status using the drug formulary data. We find that Provincial Catalog B drugs' revenue share increased from 15% (2013–2016) to 22% (2017–2019),[8] indicating significant market implications of Provincial

[7] The included cities are Beijing, Changchun, Changsha, Chengdu, Chongqing, Fuzhou, Guangzhou, Hangzhou, Harbin, Jinan, Nanjing, Shanghai, Shenyang, Shenzhen, Shijiazhuang, Taiyuan, Tianjin, Wuhan, Xi'an, and Zhengzhou.

[8] National Catalog B shares declined from 65% to 54%, while uncovered drugs rose from 8% to 16% and National Catalog A fell from 12% to 8%.

Catalog B design.

We also collect supplementary data on firm and drug characteristics, as well as information on government officials, which is discussed when used in empirical analysis.

4 Evidence of Local Firm Distortion in Formulary Design

In this section, we explore whether provincial governments' formulary decisions are driven by the presence of local drug producers, controlling for local demand.

4.1 Baseline Drug-Level Analysis

We conduct a cross-sectional comparison to test the existence of distortion in provincial formularies. We use the baseline sample at the drug-province level in the 2009 and 2017 formulary versions to estimate the following regression equation:

$$\mathbb{1}(ins)_{mpt} = \beta_0 + \beta_1 \mathbb{1}(local)_{mpt-1} + \gamma_m + \theta_{ptd} + \varepsilon_{mpt}, \tag{1}$$

where $\mathbb{1}(ins)_{mpt}$ indicates whether drug m was included in province p's formulary version t (2009 or 2017). $\mathbb{1}(local)_{mpt-1}$ indicates the presence of any local firm producing m in the prior year, i.e., 2008 or 2016 for the respective version. γ_m denotes drug fixed effects, and θ_{ptd} denotes province-by-version-by-disease fixed effects. Standard errors are clustered at the drug level.

One concern is that time-varying provincial characteristics may influence both local production and formulary inclusion. Another threat to identification is potential heterogeneity in disease prevalence across provinces before formulary updates (i.e., in 2008 or 2016). Differential disease burdens could affect regional demand, thereby influencing both local production and formulary inclusion. We therefore add province-by-version-by-disease fixed effects, so that identification rests on variation within drug classes treating the same disease in the same province and formulary version.

Table 1 presents the baseline results. The presence of local firms increases the probability of insurance coverage by four percentage points. The size is economically significant given that the mean of the dependent variables is 41%. Thus, drugs with pre-existing local production are more likely to gain formulary inclusion.

Appendix Table A3 shows robustness results. Equation (1) is re-estimated using (i) samples

Table 1: Local Firm and Formulary Design

	(1)	(2)	(3)
Local Firm Presence in the Year before the Formulary	0.045***	0.044***	0.043***
	(0.004)	(0.004)	(0.005)
Observations	151,362	151,362	147,744
Drug FE	Y	Y	Y
Province FE	Y	Y	Y
Version FE	Y	Y	Y
Province × Version FE	N	Y	Y
Province × Version × Disease FE	N	N	Y
R^2	0.761	0.762	0.778
Adj. R^2	0.757	0.757	0.760

Notes: This table presents baseline results using the sample of 2009 and 2017 formulary versions. Each observation is a drug by province by version. "Local Firm Presence in the Year before the formulary" indicates whether there was at least one local firm producing the corresponding drug in the year before the formulary change in 2009 or 2017. Standard errors are clustered at the drug level and shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

restricted to single versions (2009 or 2017), (ii) an extended sample combining 2005, 2009, and 2017 versions, (iii) two-way clustered standard errors at the province and drug levels, (iv) an alternative independent variable: the number of local firms, and (v) stricter controls (province-by-version-ATC4 fixed effects). Across all specifications, the results remain consistent with the baseline findings. In addition, since the drug registration process usually takes considerable time, firms are nearly unable to enter the market in the short run in response to possible information leakage regarding a pending formulary revision. Hence, reverse causality is not a concern.

4.2 Event Studies with Sales Data

The formulary distortion found above is rational only if local firms benefit from insurance coverage. To verify this, we employ a difference-in-difference design to examine how formulary inclusion affects drugs' total sales revenues and local firms' market shares.

The analysis sample consists of drugs not covered by any formulary in 2013–2016 and added to Provincial Catalog B in some, but not all, provinces in 2017. We compare drug-city observations added to Provincial Catalog B in 2017 (treatment group) with those never covered during 2013–2019 (control group). Because sales data are city-level, the comparison is essentially across cities. To avoid contamination from cross-drug substitution, we exclude from the control group any drugs whose ATC-3 therapeutic class had other drugs experiencing insurance coverage changes in 2017.

We estimate the following event-study regression:

$$Y_{mct} = \sum_{k=-16}^{16} \beta_k \mathbb{1}(t = \tau + k) \times \mathbb{1}(ins)_{mc} + \delta_{mc} + \lambda_t + u_{mct}, \tag{2}$$

where Y_{mct} denotes the outcome variable of interest for drug m in city c and quarter t—either the logarithm of sales revenues or the revenue share of local firms. τ indicates the first quarter of 2017. $\mathbb{1}(t = \tau + k)$ indicates the calendar quarter that is k quarters before or after the first quarter of 2017. $\mathbb{1}(ins)_{mc}$ indicates whether drug m is covered by Provincial Catalog B in city c in 2017. We omit the interaction term for $k = -1$ (the quarter immediately preceding the formulary change in 2017) and use it as the reference period. Drug-by-city fixed effects δ_{mc} absorb time-invariant local preferences, while year-quarter fixed effects λ_t control for national shocks. Standard errors are two-way clustered at the drug and city levels.

The coefficients β_k for $k \geq 0$ capture the dynamic effects of insurance coverage; those for $k < -1$ test for pre-trends. If formulary changes were driven by local demand shocks, we would observe differential pre-trends.

Figure 2 presents the event-study estimates. Panel A shows that the drugs added to Provincial Catalog B in 2017 experienced a significant relative increase in log revenues, rising over 50% within three years. This result confirms that insurance coverage stimulates demand. Importantly, treatment and control groups exhibit parallel pre-trends, with no sudden demand shift immediately before the formulary change—evidence that local demand shocks did not drive the formulary revision. Panel B shows that local firms' revenue share increased after formulary inclusion, with gains persisting for several years. This suggests that local firms benefit disproportionately from insurance coverage, likely through sustained marketing and detailing efforts.

4.3 Mechanism: Government Incentives to Favor Local Firms

In this subsection, we examine two potential incentives for provincial officials to favor local firms in formulary design: political promotion and collusion. We measure these incentives using biographical data on top provincial leaders (party secretaries and governors)—specifically their ages, places of birth, and dates of appointment—linked to the timing of provincial formulary

Figure 2: Event Study of the 2017 Provincial Formulary Change

(a) Panel A. Log Revenues

(b) Panel B. Local Firms' Revenue Share

Notes: This figure plots the effect of adding drugs to insurance coverage on drugs' total sales revenue (Panel A) and local firms' revenue share (Panel B). The analysis sample consists of drugs not covered by any formulary in 2013–2016 and added to Provincial Catalog B in some, but not all, provinces in 2017. Each observation is a drug-city-quarter. We compare drug-city observations added to Provincial Catalog B in 2017 (treatment group) with those never covered during 2013–2019 (control group). To avoid contamination from cross-drug substitution, we exclude from the control group any drugs whose ATC-3 therapeutic class had other drugs experiencing insurance coverage changes in 2017. Standard errors are two-way clustered at the drug and city levels. Dashed lines indicate 95% confidence intervals.

revisions. While formulary design involves multiple government branches and reflects the interests of various departments, top provincial leaders possess ultimate political authority and can shape final decisions even without direct involvement.

Promotion Incentives. Political economy literature links officials' promotion prospects to age (Fang, Li and Wu, 2022; He, Wang and Zhang, 2020). We therefore create a dummy equal to one if either the provincial party secretary or governor was aged 59 or younger at the time of the formulary revision, indicating they were not in their final term and thus faced stronger promotion incentives (Persson and Zhuravskaya, 2016). Interacting this variable with the local firm indicator, Table 2 Panel A column (1) shows that stronger promotion incentives are associated with a 3 percentage-point increase in local favoritism.

Collusion Incentives. We proxy for capture by local interest groups using a dummy indicating whether any top provincial official at the time of revision was born in the province, which prior literature associates with higher collusion likelihood (Jia and Nie, 2017). Panel A column (2) reveals greater local favoritism in provinces with native officials.[9]

[9]The native measure may also reflect officials' information advantage. To rule out this possibility, we interact drug quality measures with the native dummy and find no evidence that native officials are more likely to

Table 2: Heterogeneity Analysis

	(1) Officials' Age	(2) Officials' Nativity	(3) Firms' Ownership
Local Firm Presence in the Year before the Formulary	0.019***	0.038***	
	(0.007)	(0.005)	
Local Firm Presence in the Year before the Formulary × $\mathbb{1}(Age < 60)$	0.030***		
	(0.008)		
Local Firm Presence in the Year before the Formulary × Native		0.013***	
		(0.007)	
Local SOE Firm Presence in the Year before the Formulary			0.067**
			(0.026)
Local JV Firm Presence in the Year before the Formulary			0.043***
			(0.008)
Local Private Firm Presence in the Year before the Formulary			0.037***
			(0.005)
Observations	147,744	147,744	147,744
Drug FE	Y	Y	Y
Province × Version × Disease FE	Y	Y	Y
R^2	0.778	0.778	0.778
Adj. R^2	0.760	0.760	0.760

Notes: This table presents heterogeneity analysis using the baseline sample (2009 and 2017 formulary versions). Each observation is a drug-province-version. The outcome is a dummy indicating formulary inclusion. Columns (1) and (2) examine provincial officials' characteristics. "Age < 60" indicates whether the party secretary or governor was aged 59 or younger at the time of the formulary revision. "Native" indicates whether any top leader was born in the province. Column (3) examines local firms' ownership types. Standard errors are clustered at the drug level and shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.4 Mechanism: Heterogeneity by Firm Characteristics

In this subsection, we explore potential mechanisms of local preferences by examining firm characteristics.

Ownership Type. We classify local firms into three ownership types—SOE, JV, and private firms—and construct subsamples.[10] Each subsample consists of all non-local drug-province observations and local observations with producers of a single ownership type. Table 2 Panel B shows that local favoritism is strongest for SOEs, with a 6.7 percentage point increase in local favoritism, compared to 4.3 percentage point for JVs and 3.7 percentage point for private firms. These results align with Barwick, Cao and Li (2021), who document similar patterns in China's automobile industry. The differential response by ownership type supports the local protectionism interpretation: if demand shocks drove the correlation between formulary

exclude low-quality drugs—especially those whose quality information is not nationally revealed at the time of revision—from the formulary. See Section 4.5 for more details.

[10] We exclude foreign firms as none are local.

inclusion and local firm presence, all ownership types should respond similarly.

Economic Contributions. Local firms contribute to the local economy through tax revenues and employment. We therefore examine whether local governments favor firms with greater tax contributions and employment. Using data from the Chinese State Administration of Tax (SAT) survey and the Annual Survey of Industrial Firms (ASIF), we construct subsamples. Each subsample consists of all non-local drug-province observations and local observations with producers' total tax payments or employment above or below the median.[11] Table A4 shows that local governments favor local firms with higher tax contributions (columns (1)–(2)), greater employment (columns (3)–(4)), and more R&D activities, measured as conducting clinical trials (columns (5)–(6)).[12] These measures may be correlated with each other and with firm size, so results should be interpreted cautiously.

4.5 Quality Distortion

Having established that local firm presence influences formulary design, we now examine whether this favoritism affects social welfare by testing whether locally favored drugs are of lower quality. We draw on two quality data sources.

Quality Inspection Data. Provincial medical product administrations regularly conduct random inspections and publish quality reports. We collect reports from all provinces during 2005–2023 and construct dummies indicating whether local and non-local products had reported quality issues in provinces other than their home or formulary province.[13] Approximately 10.2% of drug products had reported quality issues.

We estimate the following regression equation:

$$\begin{aligned} \mathbb{1}(ins)_{mpt} = \beta_0 + \beta_1 \mathbb{1}(local)_{mp,t-1} + \beta_2 \mathbb{1}(local\_prob)_{mpt} \\ + \beta_3 \mathbb{1}(nonlocal\_prob)_{mpt} + \gamma_m + \theta_{ptd} + \epsilon_{mpt}, \end{aligned} \tag{3}$$

where $\mathbb{1}(local\_prob)_{mpt}$ and $\mathbb{1}(nonlocal\_prob)_{mpt}$ indicate pre-existing quality problems for local and non-local products prior to the formulary revision. To capture the possibility that some quality issues may affect formulary design even before its official reporting in the data, we

[11] We measure total firm-level tax payments as product-level tax data are unavailable.

[12] Results are robust to alternative tax measures and datasets (Table A5).

[13] Excluding home province reports mitigates conflict-of-interest concerns. Common reported quality problems include product traits, identification, and content determination not meeting relevant standards.

also construct indicators using all reports through 2023 to verify robustness. Standard errors are clustered at the drug level.

Table 3 Panel A shows the results. Column (1) focuses on quality problems reported in 2008 (or 2016) or earlier, while column (2) includes all reports during 2005–2023. The results show that local firm presence increases formulary inclusion by 1.7–1.9 pp. Quality problems among non-local products reduce inclusion by 1.9–2.0 pp,[14] but quality problems among local products have no significant negative effect.

Adverse Reaction Reports. The National Medical Products Administration periodically publishes reports identifying drugs with severe adverse effects or safety issues. We construct a dummy indicating whether a drug had such issues reported prior to formulary revision; approximately 5.7% of drugs in our sample did.

We estimate the following regression equation:

$$\begin{aligned} \mathbb{1}(ins)_{mpt} = \beta_0 + \beta_1 \mathbb{1}(local)_{mpt} + \beta_2 \mathbb{1}(LowQuality)_{mt} \\ + \beta_3 \mathbb{1}(local)_{mpt} \times \mathbb{1}(LowQuality)_{mt} + \gamma_m + \theta_{ptd} + v_{mpt}, \end{aligned} \tag{4}$$

where $\mathbb{1}(LowQuality)_{mt}$ indicates pre-existing reported severe adverse reactions or safety issues prior to the formulary revision year.

Panel B shows similar results to Panel A. Adverse reactions reduce formulary inclusion probability, but this effect is attenuated (or even reversed) for drugs with local producers. These results suggest that local favoritism may lead to insurance coverage of low-quality products, potentially harming consumer and social welfare.

Information Advantage. We explore whether native officials—shown in Section 4.3 to favor local firms more—do so because they possess better information about local product quality. If so, they would be less likely to include low-quality local drugs, particularly those whose quality issues were not yet publicly known at the time of the formulary decision. We

[14] This negative association suggests our quality measures capture real quality rather than utilization frequency.

Table 3: Drug Formulary Design and Drug Quality

	Quality Problems Reported Before Formulary Revision (1)	Quality Problems Ever Reported in 2005–2023 (2)
Panel A. Quality Inspection Data		
Local Firm Presence in the Year before the Formulary	0.017***	0.019***
	(0.005)	(0.006)
Quality Problems of Local Products	0.009*	0.007
	(0.006)	(0.005)
Quality Problems of Non-Local Products	-0.020**	-0.019**
	(0.008)	(0.008)
Observations	147,744	147,744
Drug FE	Y	Y
Province × Version × Disease FE	Y	Y
R^2	0.764	0.742
Adj. R^2	0.729	0.711
Panel B. Adverse Reaction Reports		
Local Firm Presence in the Year before the Formulary	0.020***	0.019***
	(0.008)	(0.007)
Quality Issue Reported	-0.008*	-0.007*
	(0.005)	(0.004)
Quality Issue Reported × Local Firm Presence	0.010**	0.008**
	(0.004)	(0.003)
Observations	147,744	147,744
Drug FE	Y	Y
Province × Version × Disease FE	Y	Y
R^2	0.739	0.706
Adj. R^2	0.701	0.685

Notes: This table examines quality distortion from local protectionism using the baseline sample (2009 and 2017 formulary versions). Each observation is a drug-province-version. The outcome is a formulary inclusion dummy. Panel A uses quality inspection data; Panel B uses adverse reaction reports. Standard errors are clustered at the drug level and shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

estimate equation:

$$
\begin{aligned}
\mathbb{1}(ins)_{mpt} = \beta_0 &+ \beta_1 \mathbb{1}(local)_{mpt} + \beta_2 Native_{pt} + \beta_3 (LowQuality)_{mt} \\
&+ \beta_4 \mathbb{1}(local)_{mpt} \times (LowQuality)_{mt} + \beta_5 \mathbb{1}(local)_{mpt} \times Native_{pt} \\
&+ \beta_6 (LowQuality)_{mt} \times Native_{pt} + \beta_7 \mathbb{1}(local)_{mpt} \times (LowQuality)_{mt} \times Native_{pt} \\
&+ \gamma_m + \theta_{ptd} + \eta_{mpt},
\end{aligned} \tag{5}
$$

where $Native_{pt}$ indicates whether the top provincial official in the year before formulary version t revision was native to province p, and $(LowQuality)_{mt}$ indicates whether the drug was reported with severe adverse reactions nationally after version t revision. A negative and significant β_7 would support the private information hypothesis. However, Appendix Table A6 shows that β_7 is insignificant, inconsistent with the private information hypothesis.

5 Policy Implications

Our evidence indicates that local firms influence provincial formulary design in China. Because coverage decisions are not based solely on cost-effectiveness, this distortion can reduce welfare and help explain the 2019 shift to a unified national formulary. We use a simple framework to clarify the welfare trade-offs between provincial and national design.

Model Setup

Model Setup Consider provinces $p = 1, \ldots, P$ and diseases (or drugs) $d = 1, \ldots, D$. Let $\boldsymbol{c} = \{c_{dp}, d = 1, 2, ..., D, p = 1, 2, ..., P\}$ denote formulary choices, where c_{dp} indicates whether disease d is covered in province p. Let net surplus from coverage be $v(\boldsymbol{c}, \boldsymbol{\varepsilon})$, a function of $\boldsymbol{c}$ and random shocks $\boldsymbol{\varepsilon}$. The central government knows only the ex-ante distribution of $\boldsymbol{\varepsilon}$, while provinces observe ex-post realizations. Define ex-ante welfare as

$$W(\boldsymbol{c}) = \mathbb{E}\left[v(\boldsymbol{c}, \boldsymbol{\varepsilon})\right]. \tag{6}$$

We compare four policy regimes:

First Best Formulary, $\boldsymbol{c_0}$. A social planner observes realized $\boldsymbol{\varepsilon}$ and solves

$$\max_{\boldsymbol{c}} v(\boldsymbol{c}, \boldsymbol{\varepsilon}). \tag{7}$$

Hence $W(\boldsymbol{c_0}) = \mathbb{E}\left[\max_{\boldsymbol{c}} v(\boldsymbol{c}, \boldsymbol{\varepsilon})\right]$.

Centralized Provincial Formulary, $\boldsymbol{c_1}$. If the central government chooses province-specific formularies without observing realized shocks, it solves

$$\max_{\boldsymbol{c}} \mathbb{E}\left[v(\boldsymbol{c}, \boldsymbol{\varepsilon})\right]. \tag{8}$$

Thus $W(\boldsymbol{c_0}) \geq W(\boldsymbol{c_1})$ due to more accurate information.

Centralized National Formulary, $\boldsymbol{c_2}$. With a uniform national formulary, the central government solves equation (8) subject to $c_{d1} = c_{d2} = ... = c_{dp} = ... = c_{dP}, \forall d$. Let $\boldsymbol{c_2}$ be the solution. By construction, $W(\boldsymbol{c_2}) \leq W(\boldsymbol{c_1})$, because $\boldsymbol{c_2}$ is a result of a constrained optimization of the same objective function as $\boldsymbol{c_1}$.

Decentralized Provincial Formulary, $\boldsymbol{c_3}$. If provinces choose formularies, they incorporate private incentives Δ:

$$\max_{\boldsymbol{c}} \mathbb{E}\left[v(\boldsymbol{c}, \boldsymbol{\varepsilon}, \Delta)\right]. \tag{9}$$

By construction, $W(\boldsymbol{c_0}) \geq W(\boldsymbol{c_3})$.

The welfare difference between national uniform formulary, as in the case after 2019 ($\boldsymbol{c_2}$), versus decentralized provincial formularies ($\boldsymbol{c_3}$) decomposes as:

$$W_2 - W_3 = \underbrace{W_2 - W_1}_{\substack{\text{loss due to} \\ \text{neglect of} \\ \text{heterogeneous} \\ \text{preference}}} + \underbrace{W_1 - W_0}_{\substack{\text{loss due to} \\ \text{incomplete} \\ \text{information}}} + \underbrace{W_0 - W_3}_{\substack{\text{benefits of} \\ \text{removing} \\ \text{distortion}}}.$$

$W_2 - W_1$ represents the welfare loss from neglecting preference heterogeneity in a uniform formulary. $W_1 - W_0$ captures the welfare loss due to the central government's incomplete information about ε_{dp}. $W_3 - W_0$ reflects the benefit of removing distortions of local favoritism induced by private incentives from provincial governments. The model highlights the centralization trade-off: removing local distortions versus sacrificing information and heterogeneous preferences advantages.

Discussion Although we cannot quantify the magnitude of each component, we provide a qualitative discussion of their relative sizes. First, our empirical results indicate that distortions due to local favoritism exist and may be sizable. Covering low-quality products due to local favoritism wastes public funds and may cause other public health issues, such as antibiotic misuse. Appendix Figure A3 shows high local shares in anti-infectives (category J) and antiparasitic products, insecticides, and repellents (category P), suggesting that local favoritism may contribute to the overuse of antibiotics.

Second, we expect losses from incomplete information to decline over time. Many healthcare facilities in China are adopting electronic medical record systems, and insurance claims are being consolidated under the National Healthcare Security Administration, which should reduce provinces' information advantage.

Finally, the central government could avoid losses due to heterogeneous preferences by designing province-specific drug formularies. Extending the tailored formularies currently used for ethnic minority groups to all provinces could improve efficiency, though it creates equity concerns. Optimal policy depends on equity-efficiency trade-offs.

Model Caveats We abstract from interprovincial budget transfers. Let v denote benefits, τ denote coverage costs, and $\boldsymbol{B} = \{B_1, B_2, ..., B_p, ..., B_P\}$ denote provincial insurance budgets; first-best maximizes $v(c_p, \varepsilon_p)$ subject to $\tau(c_p, \varepsilon_p) \leq B_p$. This matches current institutional practice, where insurance funds are largely managed at provincial or sub-provincial levels before and after 2019. Allowing cross-province transfers is left for future work.

We also assume provincial private incentives Δ reduce aggregate welfare. While favoring local firms may raise local taxes or employment, these gains could be generated more efficiently by other firms or sectors. Given a perfect transfer payment system, the opportunity cost of distorted incentives outweighs their value; thus, the assumption is reasonable. In reality, the transfer payment system may not be designed perfectly and may induce other market distortions. These concerns are beyond the scope of our paper and are open to future research.

6 Conclusion

This paper documents how provincial governments use drug formulary design of public insurance to favor local firms. We find that provincial governments are more likely to cover drugs with at least one local manufacturer, holding demand fixed. Our stylized model highlights the trade-off between a national single formulary and provincial formularies: the former corrects distorted incentives of provincial governments but may not account for heterogeneous demand or rely on less accurate information.

One limitation of our paper is that we cannot account for all private incentives faced by provincial governments, as many are unobservable. Still, creative approaches to documenting such behaviors remain an avenue for future research.

The 2019 reform may also affect long-term pharmaceutical industry dynamics, altering firms' incentives to invest in quality improvement, production efficiency, and product portfolio upgrades. These impacts warrant future research.

Our results support recent Chinese policy changes that centralize social insurance management. Further research is needed to assess the trade-offs and welfare consequences of such centralization outside the context of drug formulary design.

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Appendix A. Supplementary Tables and Figures

Table A1: ATC First Level Names and Description

Code	**ATC Level 1 Description**
A	Alimentary tract and metabolism
B	Blood and blood-forming organs
C	Cardiovascular system
D	Dermatologicals
G	Genito urinary system and sex hormones
H	Systemic hormonal preparations, excl. sex hormones and insulins
J	Antiinfectives for systemic use
L	Antineoplastic and immunomodulating agents
M	Musculo-skeletal system
N	Nervous system
P	Antiparasitic products, insecticides, and repellents
R	Respiratory system
S	Sensory organs
V	Various

Table A2: List of Provinces with Drug Formulary Data

Province	Year		
	2005	2009	2017
Anhui	N	Y	Y
Beijing	N	Y	Y
Chongqing	Y	Y	N
Fujian	N	Y	Y
Gansu	N	Y	Y
Guangdong	N	Y	N
Guangxi	Y	Y	Y
Guizhou	N	Y	Y
Hainan	Y	Y	N
Hebei	Y	Y	Y
Henan	N	Y	Y
Heilongjiang	Y	Y	Y
Hubei	Y	Y	Y
Hunan	Y	Y	Y
Inner Mongolia	Y	Y	N
Jiangsu	Y	Y	Y
Jiangxi	Y	Y	Y
Jilin	Y	Y	Y
Liaoning	Y	Y	Y
Ningxia	Y	Y	Y
Qinghai	Y	Y	Y
Shaanxi	N	Y	N
Shandong	Y	Y	Y
Shanghai	Y	Y	Y
Shanxi	N	Y	N
Sichuan	N	Y	Y
Tianjin	Y	Y	N
Tibet	Y	Y	Y
Xinjiang	Y	Y	Y
Yunnan	Y	Y	Y
Zhejiang	Y	Y	N

Notes: This table lists the provinces with available data on Provincial Catalog B.

Table A3: Robustness Check: Alternative Regression Settings

	(1)	(2)	(3)	(4)	(5)	(6)
	2009 Version	2017 Version	2005, 2009, & 2017 Versions	2009 & 2017 Versions		
Local Firm Presence in the Year before the Formulary	0.020***	0.028***	0.040***	0.043***		0.036***
	(0.004)	(0.006)	(0.006)	(0.007)		(0.006)
Number of Local Firms in the Year before the Formulary					0.006***	
					(0.001)	
Observations	84,816	62,928	205,200	147,744	143,424	136,674
Drug FE	Y	Y	Y	Y	Y	Y
Province × Disease FE	Y	Y	Y	Y	Y	Y
Province × Version × Disease	N	N	Y	Y	Y	N
Province × Version × ATC4	N	N	N	N	N	Y
R^2	0.883	0.877	0.654	0.778	0.775	0.824
Adj. R^2	0.872	0.863	0.627	0.760	0.757	0.776
Clustered S.E.	Drug	Drug	Drug	Province-by-Drug	Drug	Drug

Notes: This table shows robust results. Columns (1) and (2) show the results using the sample in the 2009 or 2017 formulary version, respectively. Columns (3) uses the sample in the 2005, 2009 and 2017 formulary versions. Columns (4)–(7) uses the baseline sample in the 2009 and 2017 formulary versions. Each observation is a drug by province by version. In columns (1)–(3) and columns (5)–(7), standard errors are clustered at drug level, while in column (4), standard errors are clustered at the drug and province levels. Column(5) employs the number of local firms producing the corresponding drug one year before the formulary change in 2017 as the regressor of interest. Column (6) controls province by version by ATC 4 fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Formulary Design and Local Firms' Tax Contribution, Employment, and Clinical Trials Investments

	Tax Contribution Amount		Employment		Clinical Trials	
	above median	below median	above median	below median	yes	no
	(1)	(2)	(3)	(4)	(5)	(6)
Local Firm Presence	0.013***	0.005	0.015***	0.008	0.013***	0.006
in the Year before the Formulary	(0.005)	(0.006)	(0.004)	(0.005)	(0.005)	(0.005)
Observations	145,214	145,215	145,214	145,215	149,0774	141,352
Drug FE	Y	Y	Y	Y	Y	Y
Province × Version × Disease FE	Y	Y	Y	Y	Y	Y
R^2	0.721	0.703	0.744	0.703	0.729	0.709
Adj. R^2	0.690	0.679	0.715	0.696	0.707	0.685

Notes: This table uses the baseline sample in the 2009 and 2017 formulary versions to show the heterogeneous results across local firms' tax contribution amount, employment, and clinical trials investments. Hence, each observation is a drug by province by version. Columns (1) and (2) show the heterogeneous results across local firms' tax contribution amount, which is calculated as the total tax paid by all local firms producing the corresponding drug in a province; columns (3) and (4) show the results across employment, which is calculated as the total number of employees of all local firms producing the corresponding drug in a province; columns (5) and (6) show the results across clinical trials, which is calculated as the total number of clinical trials conducted by all local firms producing the corresponding drug in a province before the implementation of the formulary. Columns (1) and (3) ((2) and (4)) present the estimates in the subsample consisting of non-local drug-province observations and local drug-province observations whose producers have a corresponding characteristic above (below) the median across all observations. Columns (5) ((6)) presents the estimate in the subsample consisting of non-local drug-province observations and local drug-province observations whose producers have the corresponding characteristic. Standard errors are clustered at the drug level and shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Robustness Checks: Alternative Tax Contribution Measurement

	Tax Contribution Share		Local Tax Contribution		Income Tax Contribution from SAT		Income Tax Contribution from ASIF	
	above median	below median	above median	below median	above median	below median	above median	below median
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Local Firm Presence	0.017***	0.004	0.022***	0.007	0.015**	0.006	0.017**	0.003
in the Year before the Formulary	(0.004)	(0.003)	(0.006)	(0.005)	(0.007)	(0.008)	(0.008)	(0.007)
Observations	145,214	145,215	145,214	145,215	145,214	145,215	145,214	145,215
Drug FE	Y	Y	Y	Y	Y	Y	Y	Y
Province × Version × Disease FE	Y	Y	Y	Y	Y	Y	Y	Y
R^2	0.703	0.785	0.719	0.766	0.748	0.791	0.727	0.754
Adj. R^2	0.690	0.777	0.708	0.751	0.739	0.784	0.717	0.743

Notes: This table shows robust results using alternative tax contribution measurement. Columns (1) and (2) employ tax contribution share, which is calculated as the ratio of the tax paid by local firms producing the corresponding drug in the corresponding province divided by the tax paid by all firms in the province; columns (3) and (4) employ local tax contribution amount; columns (5) and (6) employ income tax contribution amount calculated from SAT; columns (7) and (8) employ income tax contribution amount calculated from ASIF. We use the baseline sample in the 2009 and 2017 formulary versions, and hence, each observation is a drug by province by version. Columns (1), (3), (5), and (7) ((2), (4), (6), and (8)) present the estimates in the subsample—each consisting of non-local drug-province observations and local drug-province observations whose producers have a corresponding characteristic above(below) the median across all observations. Standard errors are clustered at the drug level and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Drug Formulary Design, Quality, and Private Information

	Quality Issues Reported After Revision (1)	Quality Issues Ever Reported in 2005-2023 (2)
Local Firm Presence in the Year before the Formulary	0.039***	0.036***
	(0.007)	(0.006)
Quality Issue Reported	-0.006	-0.003
	(0.004)	(0.004)
Native	0.001	0.000
	(0.005)	(0.003)
Local Firm Presence in the Year before the Formulary	0.002	0.001
$\times$ Quality Issue Reported	(0.007)	(0.005)
Local Firm Presence in the Year before the Formulary	0.015***	0.011***
$\times$ Native	(0.001)	(0.000)
Quality Issue Reported $\times$ Native	0.002	0.001
	(0.004)	(0.002)
Local Firm Presence in the Year before the Formulary	0.001	0.001
$\times$ Quality Issue Reported $\times$ Native	(0.002)	(0.002)
Observations	147,744	147,744
Drug FE	Y	Y
Province $\times$ Version $\times$ Disease FE	Y	Y
R^2	0.811	0.793
Adj. R^2	0.789	0.776

Notes: Each observation is a drug by province by version. Adverse reaction report data is used to construct quality issue measure. Standard errors are clustered at the drug level and shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure A1: Share of Drugs with Local Producers

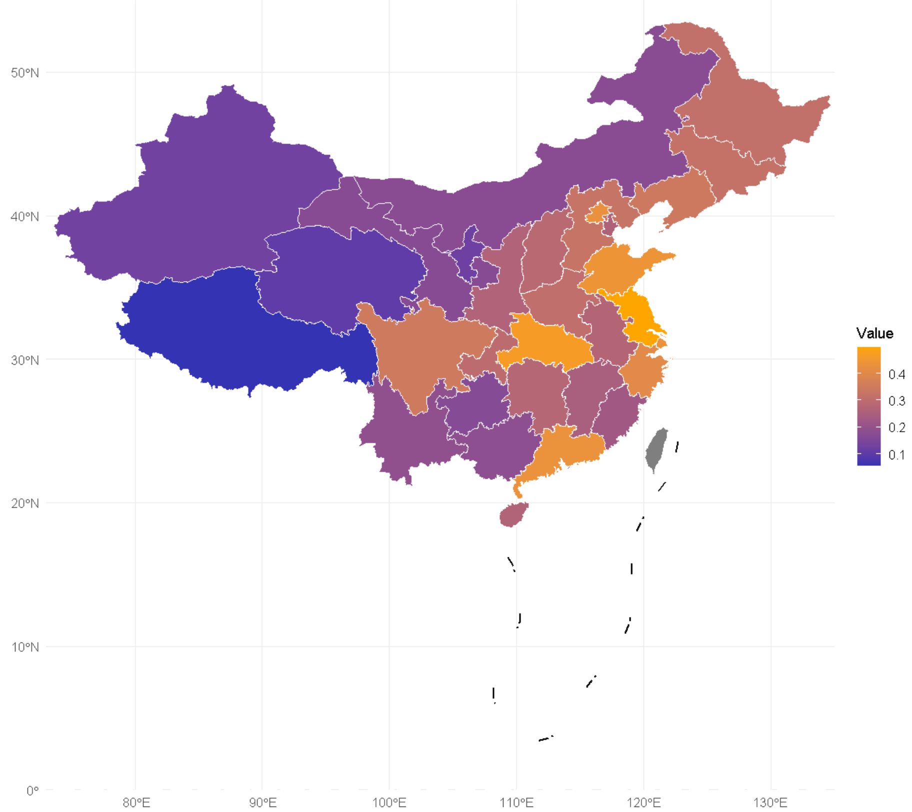
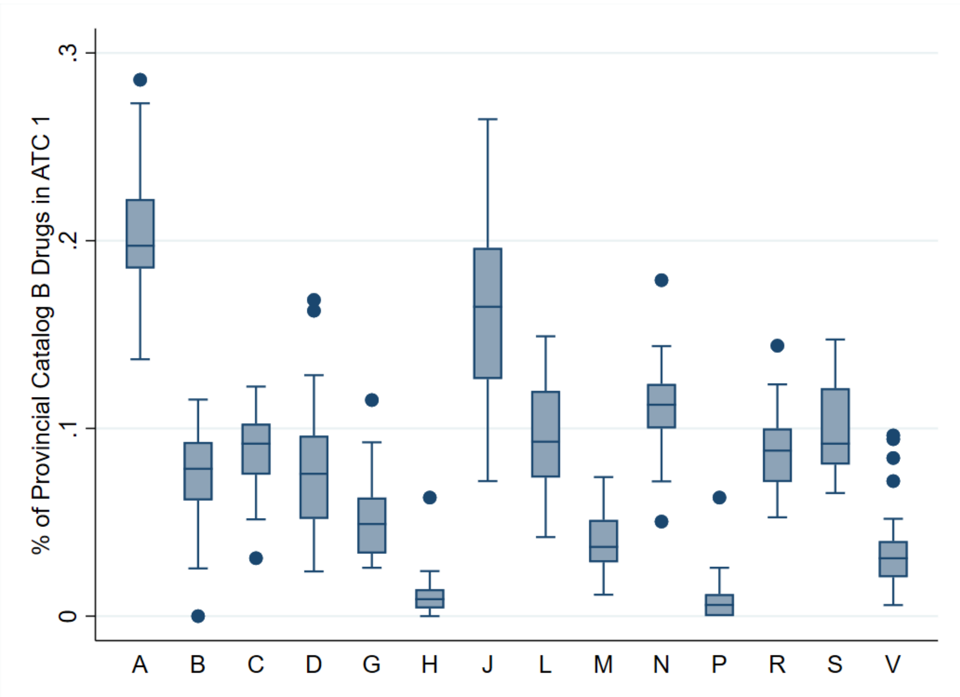
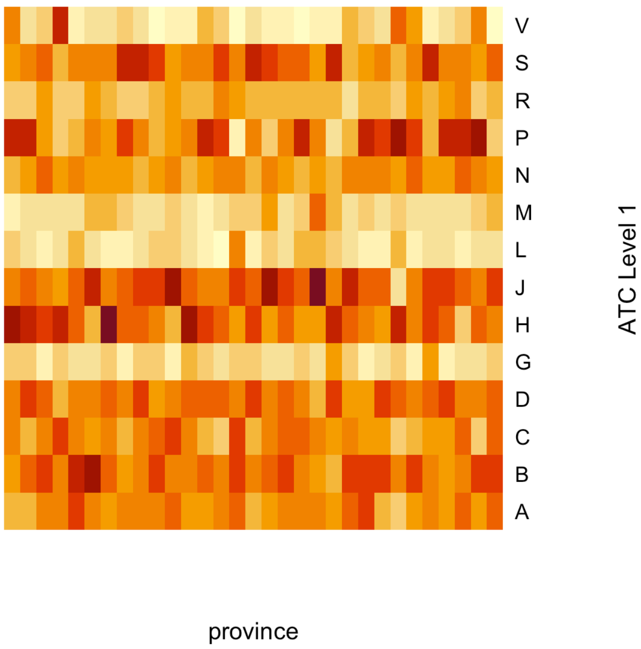


Figure A2: Distribution of the Number of Drugs Covered by Provincial Catalog B in Each ATC 1 Category across Provinces



Notes: This figure plots the distribution of the fraction of drugs covered by Provincial Catalog B in each category of ATC 1 across provinces in the 2017 formulary version.

Figure A3: Share of Local Firms by ATC Level 1



Notes: The graph shows for each province and ATC level 1, among the drugs covered by any national or provincial insurance, the share of drugs that have a local producer. The scales are normalized within each province, so the colors reflect the relative size within a province and cannot be compared across provinces.